



CATastrophic myths part 2: Common misconceptions about the environmental, nutritional, and genetic management of domestic cats and their welfare implications

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ABSTRACT

Despite the cat's popularity as a companion species, many owners and practitioners lack high quality information about important aspects of their behavior and management. Myths, anecdotes, and narratives of cats as 'low maintenance, self-sufficient' animals are pervasive, and the degree to which these may underlie complacency about fully meeting cats' needs is unknown. Several studies suggest that cat welfare and the human-cat bond may benefit from improved education about how to optimize the domestic cat's management and husbandry needs in homes and elsewhere. This paper is the second of a two-part series addressing common myths about cats. The purpose of this paper is to review and debunk common misconceptions about optimal cat care, feeding behavior, genetics, and training. Replacing these misconceptions with scientifically generated information could have a significant impact on the behavioral management of cats, positively influencing their physical health, mental stimulation, and well-being, and reducing stress for both cats and the people caring for them. Areas where further research is required to address ambiguities, and to better meet cats' needs in homes and other environments, are also identified.

Introduction

Despite their popularity as companion animals,¹ the domestic cat remains subject to misperceptions about their care, management, feeding, and behavior (Croney et al., 2023). Finding quality information on these topics can be challenging for cat owners (Pereira et al., 2014; Delgado et al., 2020), who may lack knowledge about important aspects of cat management and husbandry needs (Bir et al., 2016; Howell et al., 2016; Stella and Croney, 2016a, b; Grigg and Kogan, 2019). This paper is the second in a two-part series that aims to address common misconceptions about cats' welfare needs. It is intended to encourage reconsideration of ubiquitous, but incorrect, notions about cats, and to promote increased evidence-based discussions and recommendations that support both cat welfare and the human-cat bond.

Myth 1: Cats are low maintenance companions

A review of online discussions about 'low maintenance pets' indicates widely held beliefs that cats have minimal needs, which may contribute to their growing popularity as companions (Marston and Bennet, 2009).² However, the scientific evidence of the level of environmental consistency and complexity cats require contradicts this notion (Ellis et al., 2013; Strickler and Shull, 2014; Stella and Croney, 2016a, b; Grigg and Kogan, 2019). The disconnect between expectations and needs means that many people may not understand how to provide cats with the quality of environments and experiences they need to thrive in human homes (Stella and Croney, 2016a, b).

According to the Five Pillars of a Healthy Feline Environment (Ellis et al., 2013) every cat should minimally be provided: (1) a safe place

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¹ American Pet Products Association, 2023–2024. Pet Industry Market Size, Trends & Ownership Statistics https://www.americanpetproducts.org/press_industrytrends.asp. (Accessed 29 August 2023)

² See Also: BBC News, 2002, May 3. Britons 'prefer cats to dogs.' http://news.bbc.co.uk/2/hi/uk_news/1965499.stm. (Accessed 29 August 2023).

(such as perches, hiding areas, or access to outdoor spaces such as an enclosed cat patio, often referred to as a 'catio'); (2) safe access to key resources in their environments (for feeding, drinking, elimination, scratching, and resting); (3) opportunities to engage in play that mimics natural predatory behaviors; (4) positive and consistent human–cat social interactions; and (5) an environment that reflects the importance of the cat's sense of smell and use of olfactory communication.

It is unclear how effectively information about the Five Pillars has reached cat owners or the degree to which they are currently implemented (Stella and Croney, 2016a, b). In a recent survey-based study, only around half of cat owners reported offering their cats climbing structures, around half of the respondents offered daily interactive play, and fewer than 69% of respondents provided their cats with a dedicated scratching surface. Cat owners were more likely to report that their cat had a 'quiet, private hiding space' (92% of respondents) and objects for solo play (81% of respondents; Grigg and Kogan, 2019), suggesting that owners prioritize some aspects of cats' needs better than others.

Few cat owners report providing enrichment such as food puzzles or reinforcement-based training for their cats (Grigg and Kogan, 2019; Delgado et al., 2020). Insufficient stimulation and exercise have been associated with cat owner reports of behavior problems such as elimination outside the litter box, aggression (Ellis et al., 2013; Strickler and Shull, 2014; Grigg and Kogan, 2019) and health problems such as feline obesity - a frequently observed and often preventable condition defined as 30% above ideal body weight and equivalent to 8/9 body condition score (German, 2006).

Environmental stressors, including insufficient environmental enrichment, inadequate space, or poor social interactions with conspecifics have been associated with health-related conditions such as feline idiopathic cystitis (Buffington, 2011), digestive upset (Stella et al., 2011), and emotional overeating (McMillan, 2013). Conversely, efforts to reduce stressors, such as providing additional litter boxes, spending more time with the cat, and reducing tension between cats within the household, have been shown to improve the health and behavior of domestic cats (Buffington et al., 2006). While the relationship between environment and health outcomes can be complex, the effects of environmental modifications on outcomes lends further credence to the idea that providing cats with environments tailored to meet their individual and species-typical needs is crucial to supporting their health and welfare.

Myth 2: Cats cannot be trained

There is ample evidence that cats are trainable. A world record dedicated to the number of tricks performed by a cat in one minute currently stands at 26 tricks.³ Research has been conducted on successful cat clicker training programs in animal shelters (Grant and Warrior, 2019). Pet kittens are readily able to complete 6-week positive reinforcement-based training programs mastering behaviors ranging from 'sit' and 'down', to walking on leash and performing complex tricks (Vitale et al., 2019). Recently, a cat was shown to model human behavior, after successfully reaching the generalization stage of imitation-based 'Do As I Do' training (Fugazza et al., 2021). Training has also been successfully used with cats in cooperative care, facilitating low-stress methods for blood collection (Lockhart et al., 2013), and in

³ See: Most tricks performed by a cat in one minute. Guinness World Records, <https://www.guinnessworldrecords.com/world-records/391893-most-tricks-performed-by-a-cat-in-one-minute> (Accessed 29 August 2023).

scientific research study protocols (Thompson, 1959; Timney et al., 1985; Timney, 1990; Keeling et al., 2008; ⁴ Schapiro et al., 2018).

Nonetheless, evidence of the misconception that training is not a common or relevant aspect of cat welfare remains. For example, Salman et al. (2000) reported on characteristics of surrendered dogs and cats across 12 shelters as acquired from interviews with owners during relinquishment. While questions were asked about the training history of relinquished dogs (less than 10% had received formal training), no data on training history of cats was included. In another study surveying pet owners about cat and dog ownership and management practices, researchers found that dog training history correlated with health and welfare outcomes, however the study authors stated that 'training variables were not asked of cat owners since we did not believe them to be applicable' (Slater et al., 2008). Such implicit biases can limit availability of knowledge on the prevalence of cat training within the population and the welfare implications of training for cats.

More research on cat training protocols and their effects may also be helpful in overcoming perceptions of cats as challenging to train. Such beliefs may be partly due to insufficient knowledge of how to best motivate cats, and how to design training sessions that are compatible with a cat's behavioral ecology and lifetime experiences (Vitale et al., 2019). While general principles of animal learning theory are applicable across a wide range of species (Wynne and Udell, 2020), some training approaches may be more effective for cats. For example, shorter training sessions (instead of more traditional hour-long training classes) have been found to be beneficial when working with cats (Stella et al., 2014; Vitale et al., 2019). The welfare implications of cat training experiences are significant because such training can 1) reduce the need for handling and management practices that induce fear or stress, 2) improve behavioral success in homes, and 3) provide increased opportunities for enrichment and bonding. As the science catches up in this area, a good rule of thumb for owners should be to question the presumption that 'cats can't'.

Myth 3: Optimal feeding frequency of indoor housed cats is well-established and easy to implement

Optimizing the feeding frequency of cats has been a focus for both behaviorists and nutritional physiologists, although there is still little consensus on best practices. Research in feline nutrition has primarily focused on the effects of nutrient distribution and density on feline physiology and has largely not examined management of feeding, specifically feeding frequency, to support the health and overall well-being of indoor cats. This oversight is particularly important considering the need to balance cats' species-typical feeding patterns, their satiety, and weight management. For example, such information could help address feline obesity, a low-grade inflammatory disease (German, 2006) considered the most common domestic feline nutritional problem (Colliard et al., 2009; Courcier et al., 2010), which predisposes cats to many adverse health conditions (Scarlett and Donoghue, 1998; Lund et al., 2005). Multiple factors, such as diet composition (nutrient density or nutrients that promote energy expenditure), amount of food offered, single vs. multiple cat households leading to feeding conflict, individual cat characteristics (breed, age, body condition score, and activity level) and feeding frequency can influence food intake and nutrient utilization that create a positive energy balance and result in weight gain (German, 2006; Laflamme, 2006; German et al., 2010). There is a great need to better understand the impacts of these variables and to translate the

⁴ See: Schapiro S.J., Magden, E.R., Reamer, L.A., Mareno, M.C., Lambeth, S. P., 2018. Chapter 32: Behavioral training as part of the health care program. In: Weichbrod, R.H., Thompson, G.A.H., Norton, J.N., (Eds.). *Management of Animal Care and Use Programs in Research, Education, and Testing*. Second Edition. CRC Press/Taylor and Francis, Boca Raton, Florida USA. <https://www.ncbi.nlm.nih.gov/books/NBK500422/> (Accessed 5 September 2023).

resulting knowledge into best practices that can be utilized by those who care for cats.

With respect to feeding frequency, [Sadek et al. \(2018\)](#) recommended that cats' daily food allowances should be divided into multiple small meals fed throughout the day and that using puzzles and variation in feeding methods may further support their natural feeding behavior, improving the welfare of indoor-housed cats. However, the science behind such recommendations has been mixed. For instance, after being fed the same amount of a wet food with high caloric and moisture density once per day or four times per day for 3 weeks, cats fed four times a day had lower hormones that signal satiety (gastric inhibitory peptide [GIP] and glucagon-like peptide 1 [GLP-1]) and no differences in leptin, compared to cats fed once per day ([Camara et al., 2020](#)). In contrast, cats fed dry extruded food, had higher levels of satiety hormone leptin when fed four times daily vs. twice daily ([Deng et al., 2013](#)). The differences in the satiety hormones response suggest that food format (moisture and macronutrient content), may also be a component of satiety and should be considered in future research. Other research has suggested that feeding six vs. two meals a day does not support weight loss in cats ([Witzel-Rollins et al., 2022](#)), and a recent review that included multiple species found that greater eating frequency did not result in reduced food intake or weight loss ([Raynor et al., 2015](#)). Cats in environments with poor food availability have been found to increase meal size ([Kanarek., 1975](#)), decrease feeding frequency, and show greater agonistic behaviors towards other cats ([Ligout et al., 2020](#)).

Feeding frequency may also affect voluntary activity. Overweight cats perform less voluntary physical activity, measured using activity monitors, than lean cats ([De Godoy and Shoveller., 2017](#)), which may contribute to further weight gain. Cats fed two or four times daily vs. one time per day performed greater voluntary physical activity measured using activity monitors ([Deng et al., 2013](#); [De Godoy et al., 2015](#); [Camara et al., 2020](#)) which may support the idea that more frequent feedings could contribute to cat health; however, the nature of the specific additional activity has not been documented. Nutritional enrichment, via use of puzzle feeders or hiding of food, may offer a means to balance the need to simulate natural cat foraging and feeding behaviors and the need to keep cats at a healthy weight by decreasing the energy balance equation where energy expenditure is increased and energy intake is decreased ([Rochlitz, 2005](#); [Delgado and Dantas, 2020](#)).

A study observed that cats eat to maintain gastrointestinal bulk ([Kanarek., 1975](#)). The idea that cats eat to volume was later supported by the work of [Hirsch et al., \(1978\)](#) who found that cats reduce their intake volume when kaolin is added to a diet. Other research suggests that water may help to reduce caloric intake and support weight maintenance or promote weight loss in cats ([Wei et al., 2011](#)) and may be considered when feeding frequency and wet vs. extruded food is provided. When food is freely available, cats often eat several small meals each day ([Bradshaw, 2006](#); [Sadek et al., 2018](#)) and adjust intake depending on the bulk density, meaning the dietary fiber content is inversely related with food intake. [Bradshaw \(2006\)](#) noted that cats are opportunistic, solitary hunters, requiring them to catch and eat many small prey daily ([Bradshaw, 2006](#)). More recently, [Kohl et al., \(2015\)](#) found that wild carnivores select prey or feed on specific tissues according to the nutrient composition, further supporting the reciprocal relationship between feeding frequency and nutrient density. However, prey and commercial diets have different compositions. Indeed, mice contain animal fiber (hair) and other indigestible components such as bone ([Plantinga et al., 2011](#)), while commercial diets replace animal fiber with plant fiber, which is largely added to feline diets to reduce food intake, reduce hunger, and support gut health ([Fahey et al., 2004](#)). For these reasons, higher dietary cellulose and greater water content are often used to promote weight maintenance or loss in cats ([Prola et al., 2006](#)) and diets high in fat must be avoided or fed less frequently to avoid weight gain.

Striking a balance between meeting the behavioral and nutritional needs of cats requires recognition that in the wild, obtaining prey

requires considerable energy expenditure from cats and the animals that are consumed are quite small and of low caloric density ([Bradshaw and Thorne., 1992](#); [Bradshaw, 2006](#); [Sadek et al., 2018](#)). In contrast, cats kept as companions in homes (especially those living solely indoors) are often relatively inactive and offered highly palatable, high-energy kibble, in addition to treats ([Shoveller et al., 2016](#); [De Godoy and Shoveller., 2017](#); [Dodd et al., 2020](#)). Because of the significant differences between outdoor and indoor environments, the frequency of feeding of indoor cats needs to consider the energy density and availability of the diet. If cats will be provided with food ad libitum, feeding high fiber/low calorie diets may help to promote satiety in a free feeding environment; however, intake of high energy (high fat) diets cannot be provided ad libitum or overconsumption is likely to occur and should be control fed in larger and fewer meals.

Overall, cats fed once a day may be more satiated than those fed four or more times daily, but further research is needed to better understand the effects of frequency of feeding on cat welfare and behavior. Specifically, such research should combine physiologic measures of energy balance, consistent housing environment, and consideration of food format (as used by [Deng et al. \(2013\)](#) and [Camara et al. \(2020\)](#)) with behavioral responses to understand if these variables may alter our recommended feeding frequency in individual animals.

Myth 4: Cat coat color is associated with personality and behavioral traits

Many people believe that a cat's behavior can be predicted by their coat color ([Podberscek and Blackshaw, 1988](#); [Stelow et al., 2016](#)). For example, [Delgado et al. \(2012\)](#) found that survey participants were more inclined to characterize orange cats as 'friendlier', and to describe white and tri-colored cats as more 'aloof' than cats of other colors ([Delgado et al., 2012](#)). Given the genetic complexities that determine both coat color and behavior, predicting the latter based on the former is fraught with challenges, because each is influenced by many genes. Cat coat colors and patterns are controlled by at least ten genes and loci, some with multiple alleles, and some subject to epistasis ([Kaelin and Barsh, 2013](#)) demonstrating significant genetic complexity. While the coat is controlled almost entirely genetically, this is not the case for behavior or personality, which result from not only complex genetics, but the cat's environment, and interactions between the two ([Overall et al., 2014](#)).

Cat coat color and patterns in relation to personality, behavioral traits, tameness, aggression, likelihood of being seen at a specialty behavior clinic, and even shelter outcomes have been investigated ([Takeuchi and Houpt, 2003](#); [Delgado et al., 2012](#); [Stelow et al., 2016](#); [Wilhelmy et al., 2016](#); [Serpell., 2000](#)) but results have been inconsistent, equivocal, and entirely lacking in uniformity. Importantly, no studies to date have utilized actual molecular coat color testing. Since there are thousands of possible combinations of cat coat color and pattern, studies have typically collapsed or combined categories of coat colors, creating ambiguity. Additionally, without molecular evidence of the genotype, specific coat color alleles are unlikely to be accurately characterized. For example, a white coat color can be inherited via two different genetic mechanisms ([Kaelin and Barsh, 2013](#)), and a dominant white coat in cats may also be epistatic (i.e., masking or hiding) many or all other coat color genes ([Umbelino, 2014](#)). Since visual inspection of the coat may not relate to genotype, this also contributes to inconsistent and inaccurate results. While reputable commercial testing companies now provide feline coat color genetic testing panels, challenges remain, as some cat color loci do not yet have specific tests (e.g., orange; [Kaelin and Barsh, 2013](#)). Given these realities, animal professionals in all fields should exercise great caution in assuming behavior or personality in relation to cat coat colors and patterns; indeed, misconceptions on this topic may impact cat welfare in areas such as relinquishment, adoption potential, and more.

Myth 5: Purebred cats are less genetically healthy than random-bred cats

The pervasive myth that random-bred cats are healthier than purebred cats largely ignores complex genetic conditions, and instead focuses only on breed-specific health concerns. While purebred cats can experience specific, often single-gene, conditions due to breeding selection (i.e., inbreeding), all cats are impacted by ancient liability variants present in the entire cat population. In some modern cat breeds, intense selection for specific traits has resulted in extreme phenotypes, which can impair health and welfare. For example, Persian, Exotic Shorthair, and other cat breeds have been selected for brachycephaly, with resulting upper airway issues (Gunn-Moore et al., 2008). The brachycephalic skull is primarily characterized by a mildly to profoundly shortened muzzle. There are concomitant changes to internal upper airway structures, which together can restrict airflow. While this phenotype and its resultant negative effects on health and welfare have been well described in dogs (Ekenstedt et al., 2020) very little research has investigated this condition in cats. One survey-based report did find that increasingly brachycephalic feline facial architecture (the decreasing length of the foreface) was associated with increased respiratory noise when the cat was sleeping, breathing difficulty following exercise, and a sedentary lifestyle (Farnworth et al., 2016). Over-selection for this breed-specific anatomy has health and welfare implications, and selection practices should avoid these kinds of extreme phenotypes.

Other examples of breed-specific disease can result from recent, single-gene mutations that are isolated to a single breed or shared among related breeds. Polycystic kidney disease, observed in Persian and related breeds (Lyons et al., 2004), a single-gene, autosomal dominant nonsense mutation in *PKD1*, is a prime example. Dishearteningly, some defects are propagated intentionally despite their significant impact on welfare; for example, Manx cats, selected to be nearly or completely tailless, may also express spina bifida, urinary and fecal incontinence, rectal prolapse, and locomotor abnormalities (Gunn-Moore et al., 2008). The disproportionate dwarfism of the Munchkin cat impairs ambulation and can result in osteoarthritis and intervertebral disc disease (Buckley et al., 2020).⁵ Scottish Folds have a dominant variant that creates the forward fold of their ears, but this osteochondrodysplasia also creates painful degenerative osteoarthritis; because of this phenotype, the Scottish Fold is banned from several cat registries (Gunn-Moore et al., 2008). When available, genetic testing for breed-specific diseases should be used by breeders to avoid producing affected offspring. It is likely that only public pressure for health-conscious breeding will affect the change needed to reduce the adverse health and welfare effects of these phenotypes.

The most common genetic diseases in cats are not breed-specific, or single-gene, and are observed in all cats, including those that are random-bred. These are complexly inherited conditions where multiple genes interact with the environment. Disease examples include: feline lower urinary tract disease (inflammatory cystitis and feline urological syndrome), diabetes mellitus, and lymphocytic or plasmacytic inflammatory diseases (manifesting as gingivostomatitis or inflammatory bowel disease (Defauw et al., 2011; Harley et al., 2011; Jergens, 2012; Stella et al., 2013; Dorsch et al., 2014; Nelson and Reusch, 2014; Bell, 2016). These conditions are chronic, typically require lifelong treatment and management, and can significantly affect the cat's welfare. Disease liability genes for these complexly inherited conditions are 'ancient', meaning that they preceded breed formation and are seen in many

purebred and random-bred cats. In these cases, disease is not a result of selection, but rather the accumulation of specific liability alleles. While some purebred cats indeed may experience poor health in part because of selection for certain characteristics or unintentional concentration of deleterious genetic changes due to inbreeding, random-bred cats are by no means impervious to equally significant health problems. Veterinarians and animal welfare professionals should therefore counsel prospective cat owners about health risks that both pedigreed and random-bred cats may encounter.

Welfare implications

The propagation of the myth that cats are low-maintenance companion animals likely contributes to frustration due to unrealistic expectations for cats and cat owners alike. Veterinarians, behaviorists, shelter personnel, and companion animal educators all have important roles to play in informing the cat-owning public about the full range of cats' needs to reduce stress, prevent behavior problems, and support feline health.

Positive, consistent, and predictable human-cat social interactions are essential for cat well-being (Ellis et al., 2013). Greater recognition that cats are trainable and have the capacity to learn and problem solve within their environments (Vitale Shreve and Udell, 2015) can lead to increased opportunities for positive human-cat interactions (bonding, low-stress handling, cooperative care) and enrichment, provide opportunities for choice, and can facilitate greater access to space through leash training or ability to visit environments outside the home.

More research is needed to better understand optimal feeding practices for cats. Frequent feeding may help prevent undesirable behaviors and minimize competition for food in multi-cat households (Sadek et al., 2018). However, advice that cats should receive many small meals per day can be difficult for owners to follow and may instead be interpreted as 'cats must have food available at all times'. High individual variation also exists in cats' feeding behavior patterns (Johnson et al., 1983), making it important to tailor their feeding schedules as well as their diets accordingly (Robertson, 1999; Allan et al., 2000; Russell et al., 2000; Harper et al., 2001; Kienzle and Bergler, 2006; Colliard et al., 2009; Courcier et al., 2010; Cave et al., 2012).

Erroneous beliefs that a cat's coat color will predict behavior outcomes may have broader implications for cat welfare as well as the human-animal bond. Delgado et al. (2012) and Stelow et al. (2016) note that cats of certain coat colors may be either more sought after or at risk of not being adopted from shelters simply because of preconceptions about their behavior (e.g., that sex-linked orange female cats are more aggressive and less friendly). Thus, shelters may need to tailor the marketing of cats of different colors to offset any negative beliefs that might impede their adoption and educate the public about these misconceptions more broadly. Public awareness and education efforts should use a balanced approach, without stressing breed-specific health concerns over the frequent feline genetic health conditions, most of which appear across all breeds and mixes (Bell, 2016).

Conclusions

There is a critical need for more research on best practices in cat care, welfare, behavior, and health outcomes; there is also a need for scientists to consolidate, translate, and communicate research findings in these areas in ways that can reach professionals and broader audiences in impactful ways. While overcoming long held public misperceptions about the needs, health, and behavior of a species may not be an easy task, supporting cat welfare requires that we face this challenge head-on. This two-part series addresses just a few of the myths that must be addressed in working towards this goal. As scientists and veterinary practitioners, it is up to us to identify and provide the tools and knowledge necessary to facilitate best practices in cat care. If millions of cats are to continue to remain under the direct care of humans

⁵ See: Lyons, L.A., Fox, D.B., Chesney, K.L., Britt, L.G., Buckley, R.M., Coates, J.R., Gandolfi, B., Grah, R.A., Hamilton, M.J., Middleton, J.R., et al., 2019. Localization of a feline autosomal dominant dwarfism locus: A novel model of chondrodysplasia. *bioRxiv* <https://www.biorxiv.org/content/10.1101/687210v1> (Accessed 29 August 2023).

worldwide, we owe it to them to strive for a better scientific understanding of their biology, behavior, and needs.

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